Fixed Point Theory in Monotone Operator Theory

$A : \mathcal{H} \to 2^{\mathcal{H}}$ maximally monotone operator

$\Rightarrow$ **Resolvent $J_A = (\mathrm{Id} + A)^{-1}$ is firmly nonexpansive**

Every fixed point of J_A is a zero of A

Nonexpansivity: $\|J_A(x) - J_A(y)\| \leq \|x - y\|$

Firmly nonexpansive: $\|J_A(x) - J_A(y)\|^2 + \gamma\|x - y - (J_A(x) - J_A(y))\|^2 \leq \gamma\|x - y\|^2$

Fixed points: $\mathrm{Fix}(J_A) = \mathrm{zer}(A)$

Applications:

- *Proximal Point Algorithm:* $x_{n+1} = J_A(x_n)$
- *Douglas-Rachford Splitting:* $\mathrm{Fix}(T_{A,B})$ *solves* $0 \in Ax + Bx$
- *Forward-Backward Splitting: Variational inequalities*